

**Bangladesh University of Professionals (BUP)**

**Department of Computer Science and Engineering (CSE)**

# **SDLC Model Selection Report**

**Project: Khet-Khamar**

**Course Name:** CSE 3206: Software Engineering LAB

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## Analyze the Problem

- **Identify system requirements:** The application requires a login system using a mobile number and password, avoiding the need for an email address. The platform needs to combine a social feed for visual crop diagnosis, an agricultural marketplace, and sections for weather and news. Furthermore, the interface must support a toggle switch to change all text between English and Bangla instantly.
- **Determine project scope:** The project focuses on building a core digital community to connect rural farmers with certified experts and local buyers or sellers. We are keeping the scope manageable by focusing on offline marketplace transactions by making vendor phone numbers visible instead of building an online payment gateway.
- **Understand user needs:** Our primary users, the farmers, need a highly accessible and responsive web platform optimized for mobile browsers. Because they need to know if a vendor is trustworthy, the system must include an aggregated average star rating and a chronological list of reviews for users.

## Evaluate Project Characteristics

- **Requirement stability:** Low to Medium. As an academic project developed by an undergraduate student team, our initial ideas for the user interface and database design are very likely to change based on continuous testing and feedback.
- **Risk level:** Medium. The main risks involve the potential failure of free third-party REST APIs for weather and news. There is also a risk of scams or misinformation in the marketplace and social feed. To mitigate this, we need strict manual verification by admins to distinguish standard users from verified experts or government vendors.
- **Complexity:** Medium. Our technical implementation requires us to automatically compress image uploads to a maximum file size of 500KB to conserve local storage and bandwidth.
- **Delivery timeline:** Phased. The project delivery relies on completing different stages of the system and presenting them as an evolving working system during project demonstrations.

## Compare SDLC Models

A comparative analysis of different SDLC models was conducted based on the specific constraints of the Khet-Khamar project. Each criterion was assigned a priority score based on its importance to our system (Total Score = 30).

Table 1: **Comparison matrix with different models**

| Priority         | Criteria                      | Waterfall | Iterative | Agile     | Prototype |
|------------------|-------------------------------|-----------|-----------|-----------|-----------|
| 4                | Clear upfront requirements    | 4         | 0         | 0         | 0         |
| 6                | Early user UI testing         | 0         | 6         | 6         | 6         |
| 6                | Easy to adapt to changes      | 0         | 6         | 6         | 6         |
| 5                | Good for zero-budget projects | 0         | 5         | 0         | 5         |
| 5                | Handles complex integrations  | 0         | 5         | 5         | 5         |
| 4                | Lowers usability risk         | 0         | 0         | 0         | 4         |
| <b>Total: 30</b> | <b>Overall Score</b>          | <b>4</b>  | <b>22</b> | <b>17</b> | <b>26</b> |

## Select Suitable Model

Based on the detailed matrix analysis, we have chosen the **Prototype** model as the best methodology for the Khet-Khamar system.

## Justify Selection

The final evaluation scores from our matrix are: Waterfall (4), Agile (17), Iterative (22), and Prototype (26).

The Prototype model easily achieved the highest score because it directly supports our biggest project needs. The Waterfall model scored poorly because it does not let us change our minds easily or test the UI early. Agile is popular in the industry, but it requires a lot of structured meetings and experience that our student team does not have right now.

Since Khet-Khamar targets rural farmers, our biggest challenge is making sure the app is actually easy for them to use. The Prototype model is perfect because it allows us to build a simple, working version of the user interface first. We can show this early prototype to stakeholders to see if the Bangla language toggle and crop image uploads make sense to them. Furthermore, since we are operating in a zero-budget environment for academic purposes, building the project in prototype stages ensures that our core authentication and database structures work perfectly before we attempt more difficult tasks like caching external APIs.

Khet-Khamar - Detailed Use Case Diagram

